

Laser keyhole welding of dissimilar metals with spiral contours: Metal mixing, microstructure, and mechanical strength

Guanjin Yan^{a,*}, Masoud M. Pour^{b,*}, Teresa J. Rinker^b, Junjie Ma^b, Blair E. Carlson^b, Wenda Tan^{a,*}

^a Mechanical Engineering Department, University of Michigan, Ann Arbor, MI 48109, USA

^b General Motors Global Research & Development Center, Warren, MI 48092, USA

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ABSTRACT

Laser keyhole welding of dissimilar metals has broad applications in various industrial sectors, but, like many other fusion welding techniques, suffers from the formation of intermetallic compound (IMC) phases, which are brittle and can significantly compromise the mechanical performance of the joints. This is a critical challenge for the adoption of laser welding techniques by industry. This study focused on laser keyhole welding of lap joints of Aluminum (Al) on Copper (Cu) using spiral contours, which were expected to offer longer weld lengths in limited space and could potentially increase the maximum loading of the joints. Experiments were performed to produce spiral welds using different processing parameters and characterize the chemical composition, microstructure, and mechanical strength of the joints. Analysis revealed that by increasing the spiral distance and/or decreasing the laser power, the joint geometry is changed and the average Cu concentration in the joints is reduced, less brittle IMCs are formed in the joints, and the mechanical strength of the joints is improved. Furthermore, computational fluid dynamics simulations were leveraged to understand the dominating physics that drove the asymmetric fluid flow and metal mixing in melt pool with a curved contour, and the details of re-melting and Al-Cu re-mixing in the melt pools of adjacent spiral arcs were investigated.

1. Introduction

Laser welding of dissimilar metals has been widely used in automotive, aerospace, and other industrial sectors [1–4]. Compared with other welding methods (e.g., arc welding, friction stir welding, etc.), laser welding presents multiple advantages such as high welding speed, non-contact, and low energy input (resulting in narrow heat-affected zone and low thermal distortion). These features can provide better mechanical properties for the joints [5–9]. However, the formation of intermetallic compounds (IMCs) in laser welding (as in all fusion welding processes) of dissimilar metals is a critical challenge [2,10–15]. IMCs are generally brittle phases with low electrical conductance that can significantly undermine the mechanical and electrical performance of the joints [16]. It is important to reduce the formation of IMCs and their negative impacts on the joints.

Numerous experimental works have been conducted to investigate laser welding of dissimilar metals [10,14,16–27]. Fortunato et al. performed linear and wobbling laser scan paths on Aluminum (Al)-on-

Copper (Cu) and Cu-on-Al lap joints. They found that a linear scanning strategy produced joints with higher maximum loads compared to a wobbling scanning strategy with the same energy density [10]. Schmanel et al. investigated laser braze-welding of Al-Cu lap joints, and found that the joint mechanical performance could be improved by minimizing the material intermixture and reducing the IMC layer thickness [16]. Shin et al. used a pulsed laser to weld Al ribbon to Cu sheet and confirmed that the IMCs could reduce the mechanical strength and increase the electrical resistance when the weld area was enlarged [17]. Beyond only the phase composition, Huang et al. revealed the relationship between welding parameters, metal mixing, and IMC formation in Al-on-Cu lap joints using linear laser scans [18]. Furthermore, his team found that laser oscillation could be used to tune the chemical composition, and hence the microstructure and mechanical properties of the joints [21]. Trinh et al. studied the dendritic growth of IMCs toward Al in both Al-on-Cu and Cu-on-Al lap joints using a pulsed laser. They explained this phenomenon by the solubility limit and differences in the diffusion rate of the base metals [22].

* Corresponding authors.

E-mail addresses: masoud.mohammadpour@gm.com (M.M. Pour), wendatan@umich.edu (W. Tan).

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Compared with linear welding contours, spiral welding contours can accommodate longer welds within the same area. If a linear weld and a spiral weld are produced with the same laser power and welding speed, it can be assumed that the weld strength (i.e., the allowable mechanical load on a unit length of the weld) of the two welds is approximately the same. In that case, the total allowable mechanical loads are linearly proportional to the total lengths of the welds, and the spiral welds can provide significantly higher allowable load than the linear welds. There have been previous investigations on laser welding with spiral contours [28–36]. Cheng et al. studied spiral welding of Al-on-Cu and Cu-on-Al lap joints using a nano-second pulsed laser. For Al-on-Cu the welds were “V-shape” while for Cu-on-Al the welds were “nail shape”. They found that the Cu-on-Al configuration presented higher tensile strength than the Al-on-Cu configuration because the “nail shape” welds delivered more bounding area than the “V-shape”. They also characterized the hardness of different phases and found the hypo-eutectic phase had the lowest hardness and led to improved joint ductility [28]. Norouzian et al. performed spiral laser welding of Cu-on-Al lap joints and investigated the effects of laser absorptivity (modified by surface treatment) and laser energy input on the microstructure and hardness of the fusion zone. They also established a convolutional neural network to detect different weld categories and recognize unstable welds [29]. Du et al. studied the laser micro-joining of Al-on-Cu foils. They found that by increasing the welding speed, the volume of the interlocking structure (a combination of ductile phases and IMCs) was reduced, and the average strength of the joints was decreased [30]. Shin et al. performed an experimental investigation on spiral laser welding of Al alloy. They found that the increase of contact area could significantly increase the tensile-shear load as well as the electrical resistance [31]. Li et al. also analyzed the mechanical properties and fracture morphologies of Cu-on-Al lap joints with different spiral distances using nanosecond-pulsed-laser. They found that the overlap of adjacent welds produced defects, which could be reduced by increasing the spiral distance. They also found that increasing spiral distance helped reduce the mutual diffusion between Al and Cu and hence the formation of θ -Al₂Cu IMC, leading to better mechanical properties [35].

While these experimental works have revealed the process-property relationship in spiral laser welding, the underlying physics that determine the joint microstructure and properties are not fully understood. Computational Fluid Dynamic (CFD) can be used to study the fundamental physics in this matter. Huang et al. [18,21] simulated the fluid flow and metal mixing in linear laser welding of Al-on-Cu lap joints, and showed that the recoil-pressure-driven flow causes an upward migration of Cu into the Al-rich region in the upper half of the melt pool, while the Marangoni stress on the melt pool surface produces two flow vortices that facilitated the Al-Cu mixing. Huang et al. further investigate the spiral laser welding of Al-on-Cu lap joints, and revealed that the increase of the keyhole Reynolds number can lead to the uniform increase of metal mixing in the molten pool. Deng et al. used a combination of experiment and simulation to study spatter occurrence during laser welding. They found that zinc vapor pressure can be used as an indicator of the spatter formation and melt pool fluctuation [34]. Shi et al. studied the keyhole-induced porosity in oscillation laser welding. They concluded the keyhole size will increase as the oscillation frequency and amplitude increase; the stirring effect by switching from keyhole mode to conduction mode can reduce the porosity [37]. To the best knowledge of the authors, there is a lack of modeling work investigating the spiral laser welding of dissimilar metals, especially in the view of metal mixing.

In this work, a combination of experiments and simulations was used to investigate spiral laser keyhole welding in Al-on-Cu lap joints. Welding experiments were performed with different laser powers and welding contours, and the influences of these parameters on the chemical composition, microstructure, and mechanical strength were identified. Furthermore, a CFD model was developed to simulate the melt pool flow and metal mixing in spiral welding. The simulation clearly revealed

the non-symmetric fluid flow in the melt pool and its impact on the distribution of different chemical elements in the fusion zone. It also successfully captured the partial re-melting of the previous arc welding track by the following one and the element re-mixing during the re-melting. The simulations provided unprecedented insight into the process dynamics, which ultimately determined the microstructure and property of the joints.

2. Methodology

2.1. Experiments

Fig. 1 shows the schematic Al-on-Cu lap joints of spiral welding of dissimilar metals. Pure Al (1100) sheet of 0.2 mm thickness and Cu (110) sheet of 0.5 mm thickness were used in the experiments. An IPG Ytterbium Laser System with a spot size of 115 μm on the focal plane was used in the experiments. The laser has a maximum power output of 2240 W at the wavelength of 1064 nm. A BLACKBIRD SCU-2 scanner was used to enable the spiral welding contours. The laser forward speed was fixed as 1000 mm/s and the laser power was selected as 1410 W, 1880 W, and 2240 W. Archimedean spiral was used in this study. In a polar coordinate system (r, θ), the spirals can be described by the following equation.

$$r = b \bullet \theta \quad (1)$$

Here b is the spiral distance, which is defined as the radial distance between two adjacent spiral arcs, as shown in Fig. 2. In the parametric study of this work, the spiral distance was set to 0.1 mm, 0.3 mm, and 0.5 mm in different cases. For each welding case, the contour started inside the spiral with 3 rounds outward until the radius reached 1.5 mm. In total, nine cases with three different laser powers and three spiral distances were studied in this study as shown in Table 1.

After the welding process, the samples were cross sectioned through the center of the spirals and polished for analysis. The Cu concentration in the fusion zones was characterized with energy-dispersive X-ray spectroscopy (EDS) element mapping. The morphology of the microstructures was characterized by scanning electron microscope (SEM) with backscattered electrons (BSE). Microhardness tests were performed in and around the fusion zone to identify the heat affected zone (HAZ) in the samples. The hardness was evaluated by a QATM Qness 60 A+ EVO machine with Vickers test method HV 0.025. A test series of 8 points were performed on the top region, 4 points were in the weld while the other 4 points were in the base metal, as shown Fig. 3a. Lap shear testing was performed to evaluate the strength of the joints with an INSTRON 5966 machine using a load rate 1 mm/min. Each lap shear test sample

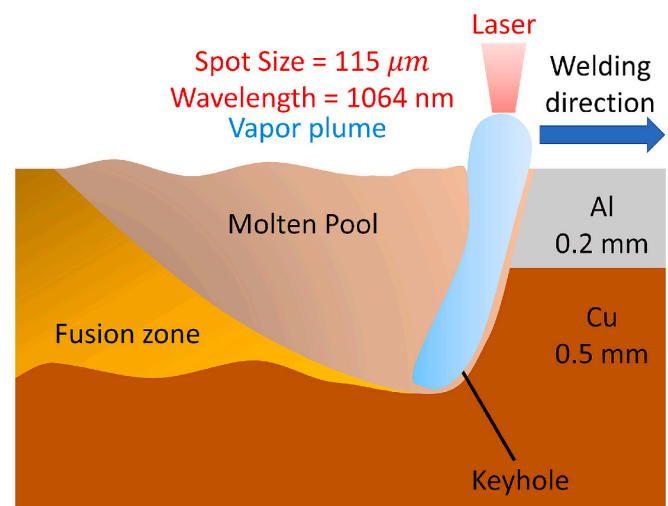


Fig. 1. Schematic of laser welding for Al-Cu lap joints.

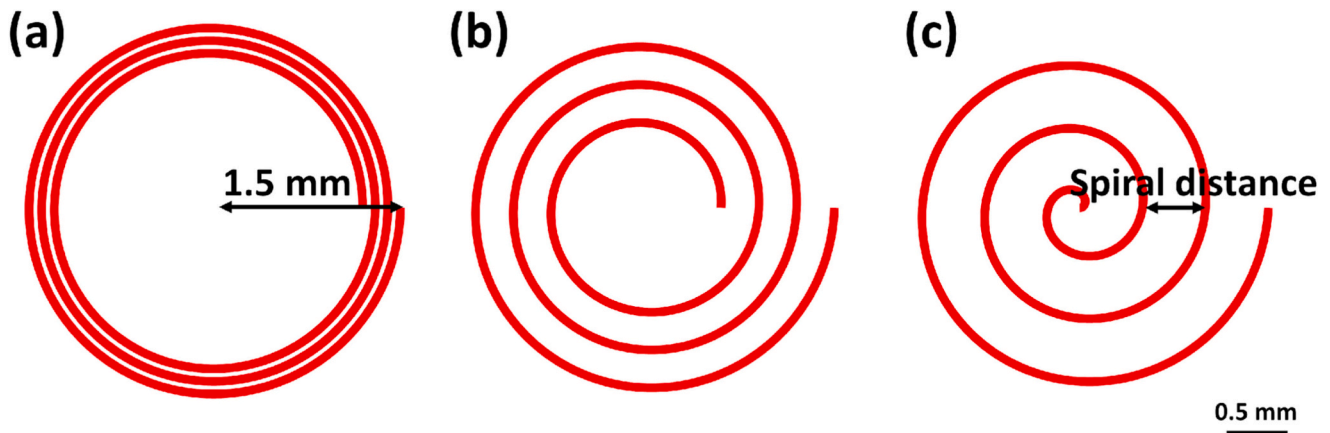


Fig. 2. Archimedean spiral and definition of spiral distance (a) spiral distance 0.1 mm, weld length = 25.45 mm (b) spiral distance 0.3 mm, weld length = 19.81 mm (c) spiral distance 0.5 mm, weld length = 14.30 mm.

Table 1

Experimental parameters ($v_{\text{laser}} = 1000 \text{ mm/s}$).

Spiral distance	0.1 mm	0.3 mm	0.5 mm
Laser power			
1410 W	#1	#2	#3
1880 W	#4	#5	#6
2240 W	#7	#8	#9

had 4 spiral welds with a spacing of 10 mm, as shown Fig. 3b.

2.2. CFD model

A 3D CFD model had already been developed in Flow3D Weld® by the team to simulate the laser keyhole welding of Al-on-Cu lap joints [18,21,23,24], and the model was used in this work to investigate the fluid flow and metal mixing in the spiral welding process. The model solved the conservation equations of mass, momentum, thermal energy, and chemical species to simulate the fluid flow, heat transfer, and element mixing in the melt pool. Volume of Fraction (VOF) [38,39] and ray-tracing methods [40,41] were used to track the dynamic evolution of keyhole and predict the laser absorption on the keyhole wall, respectively. The details of the formulation and material properties for the model have been presented in [18,21]. The model used laser parameters (power, power distribution, and scanning speed) as inputs, and predicted thermal history, melt flow field, and the redistribution of different metal elements in the melt pool.

The simulations of laser scanning over the entire spiral contours were computationally expensive. To reduce the cost, only a portion of the entire spiral was simulated for each case, as illustrated in Fig. 4(a). For the entire spiral (red curve), only the arc segments in the green dashed region were simulated. The calculation domain was set to be $4 \text{ mm} \times 3 \text{ mm} \times 1 \text{ mm}$, with two regions of different mesh sizes as shown in Fig. 4 (b). A uniform mesh size of $10 \mu\text{m}$ was used in the fine mesh region, in which the physics of keyhole and melt pool were resolved. A coarse mesh of $100 \mu\text{m}$ was applied around the fine mesh region, and only heat diffusion was simulated in this region. The timestep size was set according to the Courant-Friedrichs-Lewy convergence condition. The simulations were run on a 48-core machine with an Intel Xenon CPU 6248 R 3.00 GHz and 384 GB RAM. The simulations usually took 100–150 h.

3. Results and discussion

3.1. Chemical composition

EDS mapping was used to characterize the distribution of Cu concentration on halves of the center sections for the spiral welds. The results for cases #1–3 are shown in Fig. 5. In each case, three fusion zones, separated or connected, were generated sequentially by the three spiral arcs in the contours: the one on the left was the closest to the spiral center and produced by the first spiral arc, while the one on the right was the furthest from the spiral center and produced by the last spiral arc of the entire contour. Only two of the fusion zones were shown for case 3.

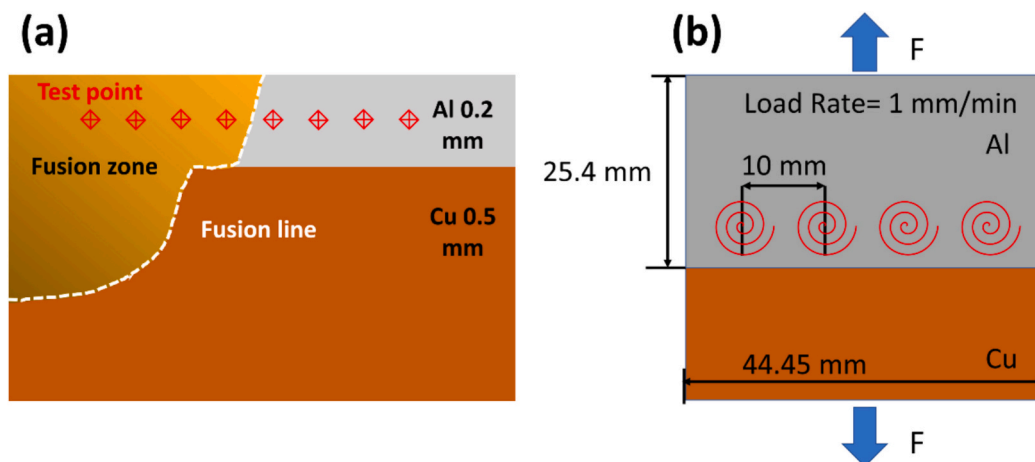


Fig. 3. (a) Schematic of microhardness test of Al-Cu lap joint's cross section (b) Top view of lap shear test setup of Al-Cu lap joints (spiral distance 0.5 mm).

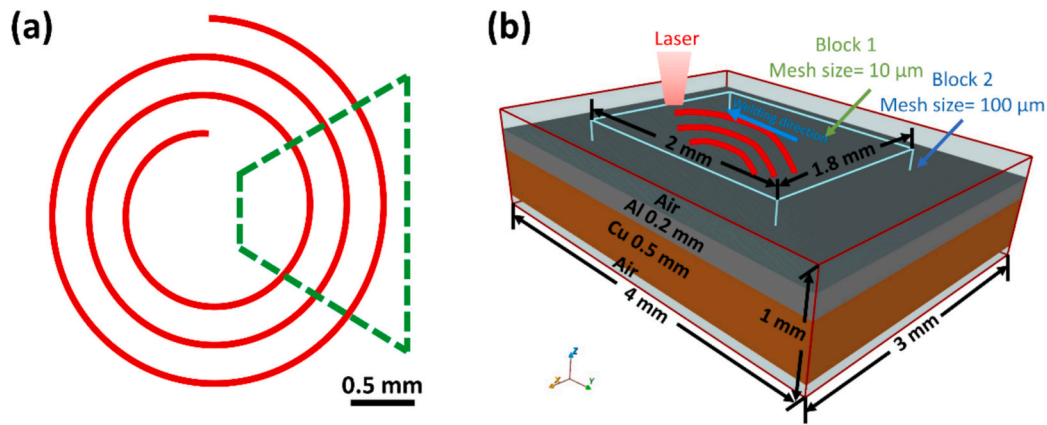


Fig. 4. Setup for computational fluid dynamics simulations. (a) Segment of spiral contour for simulation (spiral distance 0.3 mm). (b) Configuration of calculation domain.

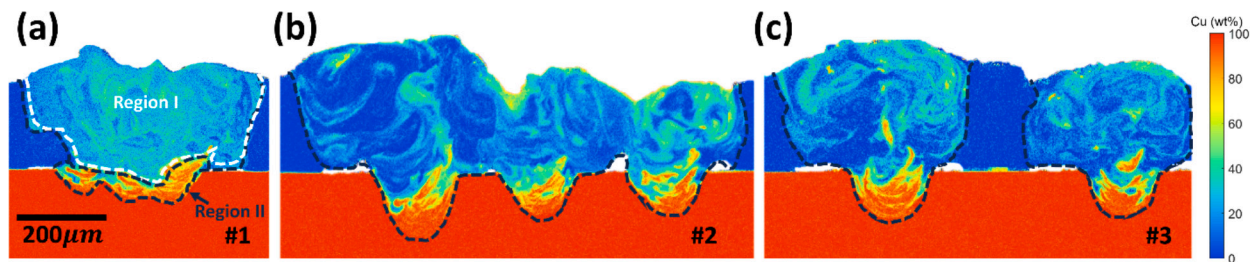


Fig. 5. EDS element mapping results of Cu concentration field for selected cases of different spiral distances. (a) case #1 (b) case #2 (c) case #3.

Blue indicated pure Al, red indicated pure Cu, and the different colors in the blue-red spectrum represented different weight percentage levels of Cu in the Al-Cu mixture.

Case #3 (Fig. 5c) was studied first as the baseline case: the fusion zones generated by the different spiral arcs were completely separated due to the large spiral distance, so each fusion zone was independent of the other ones. In each individual fusion zone, two regions of very different Cu concentrations were found. The top Al-rich region (denoted as Region I in Fig. 5a) was dominated by blue matrix (0–10 wt% Cu) with cyan strips (30–40 wt% Cu). The bottom Cu-rich region (denoted as Region II in Fig. 5a) was, on the other hand, dominated by orange-red colors (80–100 wt% Cu).

This typical pattern of Cu distribution was explained by the CFD simulation result for Case #7, as shown in Fig. 6. In spiral welding with a curved scanning contour for the laser (as shown in Fig. 6a), the general physics and the resultant flow pattern are similar to those with a linear scanning contour [18]. Fig. 6b presented a longitudinal section of the

melt pool on its center plane. As the laser illuminated the material, the metal began to melt. The liquid metal continued to absorb laser power, leading to a high evaporation rate on the molten metal surface. The resulting vapor flux generated a strong recoil pressure, pushing the molten metal away and creating a keyhole. The recoil pressure at the keyhole bottom drove an upward flow (marked by the white arrows) that brought the melted pure Cu from the bottom of the melt pool to the pure Al in the upper area. This upward flow of Cu could also be seen in Fig. 6c, which showed a cross-section of the melt pool behind the keyhole. Strong Marangoni flows were found (marked by the pink arrows) on the melt pool surface. They were driven by a combination of thermo-capillary and solute-capillary forces, and produced vortices in the melt pool that effectively stirred the Al-Cu melt to facilitate the mixing.

The effects of spiral distance were revealed by comparing cases #2 and #1 with the baseline case.

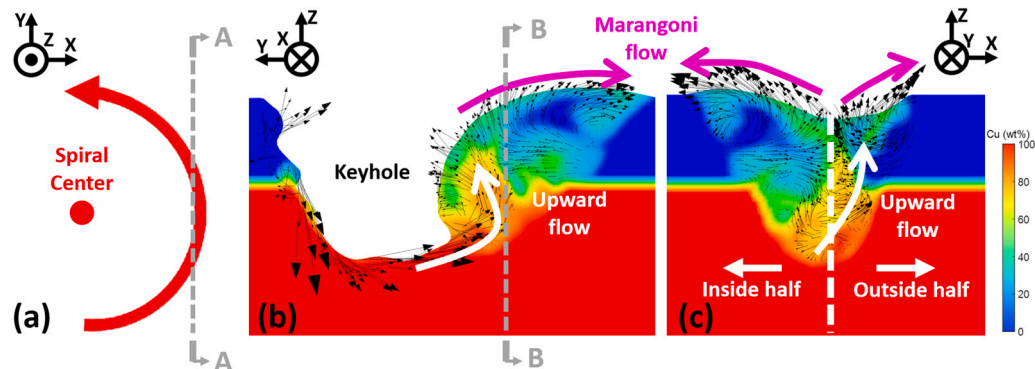


Fig. 6. Computational fluid dynamics predictions of fluid field and copper distribution. (a) Top view of the laser scanning path. (b) Simulation result on a longitudinal section labeled as A-A plane in (a). (c) Simulation results on a transverse labeled as B-B plane in (b).

- In case #2 with the spiral distance of 0.3 mm, a small portion of Region I in the previous fusion zone was re-melted by the next melt pool, and thus all three fusion zones had their Region I connected. The Cu distribution in the re-melted area of the previous fusion zone was modified by the next melt pool, and the Cu in this area was also involved in the Al-Cu mixing in the next melt pool. As the re-melted area was generally small in this case, the Cu distribution in Region I for case #2 was fairly similar to that for the baseline case. Region II of the fusion zones in case #2 were not re-melted by the next melt pools. They stayed separated and the Cu distribution in Region II was also very similar to that for the baseline case.
- In case #1, the spiral distance was further reduced to 0.1 mm. As all the three spiral arcs were so close to each other, the first fusion zone, both Region I and Region II, was re-melted by not only the second melt pool but also the third one. All the fusion zones were connected into an integrated one. The repeated re-mixing of the later melt pool significantly improved the homogeneity of Cu distribution within the integrated fusion zone.

The repeated re-melting and re-mixing were well demonstrated by a CFD simulation for Case #7. Shown in Fig. 7 was the fluid flow and Cu distribution on the center section of the spiral contour at different moment. Moment t_1 was defined as the moment when laser past the center section for the first spiral arc. At moment $t_1 + 0.0002$ s, the laser moved into the paper plane and the center section was transversely cutting through the melt pool right behind the keyhole (Fig. 7a). An upward flow could be seen on the center section (white arrow) that brought Cu to the pure Al in the upper melt pool. At moment $t_1 + 0.0004$ s, the center section was cutting through the melt pool near its tail (Fig. 7b), and the Marangoni flows were clearly seen (pink arrows) that drove the vortices to stir the Al-Cu mixture to be more uniform. At moment $t_1 + 0.001$ s, the moving melt pool had completely past the center section (Fig. 7c), and the Al-Cu mixing was complete for the first spiral arc.

The laser then came back to pass through the center section for the second time to scan the second spiral arc at moment t_2 . At $t_2 + 0.0003$ s, the center section was, again, transversely cutting through the melt pool right behind the keyhole (Fig. 7d). As the spiral distance was 0.1 mm in this case, the second melt pool re-melted large portions of Region I and Region II of the first fusion zone, both of which were already Al-Cu

mixture due to the mixing in the previous melt pool. When the upward flow was generated (white arrow), it brought up some Al-Cu pre-mixture (from the re-melted Region II of the previous fusion zone) and some pure Cu (from the first-time melt by the second melt pool) toward the upper part of the melt pool, which was also half occupied by the Al-Cu pre-mixture and half occupied by pure Al. As the laser moved forward, the center section was dominated by the Marangoni flows on the melt pool surface at $t_2 + 0.0006$ s (purple arrows in Fig. 7e), and the resultant vortices further stirred the Al-Cu mixture. The final Cu distribution in the fusion zone was significantly more uniform in the second fusion zone (Fig. 7f) than the first one (Fig. 7c). A similar process was repeated when the third spiral arc was scanned (Fig. 7g–Fig. 7i), and the final Cu distribution was also fairly uniform.

One interesting point with the curve scanning contour was that the melt pool and fluid flow therein were asymmetric, as shown in Fig. 6c. As the laser was moving along the +Y direction as well as turning toward the -X direction, the upward flow behind the rear keyhole wall was constantly turning toward the +X direction (as marked out by the white arrow). This upward tilted flow drove the Cu from the bottom plate to the top plate toward the outside half of the melt pool (i.e., the side further away from the spiral center, as labeled in Fig. 6c). As a result, the Cu concentration was slightly higher on the outside half than in the inside half of the fusion zone, as shown in the separated fusion zones in case #3 (Fig. 5c). The asymmetry was also found in case 2 (Fig. 5b), in which the fusion zones were connected. It seemed that relatively more Cu was brought to the outside half of the first fusion zone, which then, through re-melting and re-mixing, was brought to the outside half of the second fusion zone. As a similar process repeated, more Cu was ultimately brought to the third fusion zone, particularly on its outside half.

The decrease of spiral distance was also found to change the average chemical composition in the fusion zone. Specifically, as the spiral distance decreased from 0.5 mm to 0.3 mm and 0.1 mm, the average Cu concentration of the entire fusion zone, $\bar{C}_{Cu}$, was increased from 36 wt% Cu to 42 wt% Cu and then to 49 wt% Cu. In the baseline case of 0.5 mm spiral distance, $\bar{C}_{Cu}$ in the three separated fusion zones was determined by the total weights of melted Cu and melted Al in all three fusion zones. When the spiral distance was reduced to 0.3 mm, the three fusion zones were connected in their Region I. The three melt pools still melted the same amount of Cu as in the baseline case. However, the second and third melt pool had to re-melt a portion of the previous fusion zone,

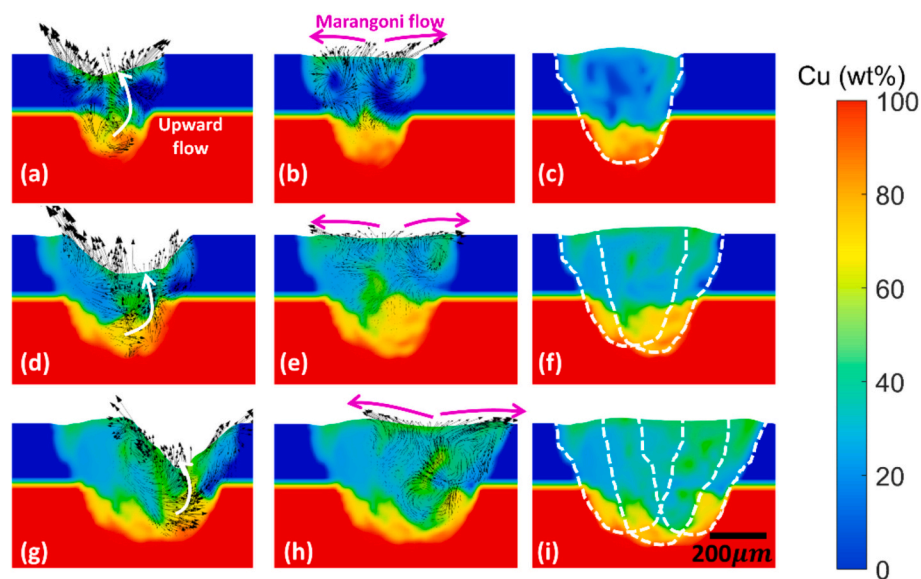


Fig. 7. Modeling predictions of fluid flow and Cu distribution on the center section for the spiral contour. t_1 , t_2 , and t_3 denoted the moment for the laser to go across the center plane for the first, second, and third time. (a)–(c) showed the simulation results at $t_1 + 0.0002$ s, $t_1 + 0.0004$ s, $t_1 + 0.001$ s. (d)–(f) showed the simulation results at $t_2 + 0.0003$ s, $t_2 + 0.0006$ s, $t_2 + 0.001$ s. (g)–(i) showed the simulation results at $t_3 + 0.0003$ s, $t_3 + 0.00054$ s, $t_3 + 0.001$ s.

which was not pure Al, so the total amount of melted Al was lower than that of the baseline case. As a result, $\bar{C}_{Cu}$ was increased in this case. Following a similar analysis for the case of 0.5 mm spiral distance, an even larger overlap between the two adjacent fusion zones led to reductions in the amounts of melted Al and melted Cu, with the Al reduction more significant than the Cu reduction. This caused a further decrease of $\bar{C}_{Cu}$ in this case.

The effects of laser power were also investigated by comparing cases #1, #4, and #7, with their EDS mapping results shown in Fig. 8. The Al-Cu mixing in the fusion zone was generally effective in these cases as little area of pure Cu or pure Al could be found. As the laser power was increased from 1410 W (case #1) to 1880 W (case #4) and then 2240 W (case #7), the fusion zone became deeper. More Cu was melted in the melt pool to be mixed with Al, so the average Cu concentration of the entire fusion zone was increased.

3.2. Microstructure

For Al-Cu lap joints, the microstructure in Region I is more critical to the mechanical properties of the joints [19] and, therefore, was selected to be the focus of the microstructure analysis for this work. From the EDS mapping results from Fig. 5 and Fig. 8, five typical chemical compositions were found within the Al-Cu lap joints, and they produced different microstructures.

- Blue regions were found to be the “background” in region I for cases #2 and #3. The blue color corresponded to the composition of 0–10 wt% Cu, which was located within the α -Al or hypo-eutectic range in the Al-Cu phase diagram [42] (blue arrow in Fig. 9a). For a α -Al composition, the solidification produced pure α -Al throughout the region. For a hypo-eutectic composition, α -Al first precipitated as the primary phase during solidification when the local temperature dropped below the liquid temperature, and the remaining liquid surrounding the primary α -Al turned into the eutectic phase when the local temperature further dropped below the eutectic temperature. The final microstructure was the primary α -Al phase surrounded by the eutectic microstructure, as shown in Fig. 9b and c.
- Cyan regions were found to be scattered strips in region I for cases #2 and #3 and dominated region I for case #1. The cyan color indicated 30–40 wt% of Cu, which was located in the hyper-eutectic region in the Al-Cu phase diagram (cyan arrow in Fig. 9a). Its solidification process should be fairly similar to that of the hypo-eutectic composition, except that the primary phase was an IMC phase θ . The final microstructure was the primary θ surrounded by the eutectic microstructure, as shown in Fig. 9d.
- Green regions were found to occupy a considerable portion of region I in case #4. The color represented the composition of 50–60 wt% of Cu, which, as marked out by the green arrow in Fig. 9a, fell around the θ region. The typical microstructure for such regions was pure θ (Fig. 9e), a brittle IMC phase that can easily crack.

- Yellow regions could be found in region I of case #7, which indicated the composition of 60–70 wt% Cu. The composition was located between the θ and η regions in the phase diagram (yellow arrow in Fig. 9a), and the microstructure is a mixture of $\theta + \eta$, two brittle IMC phases, as shown in Fig. 9f.
- Orange-red regions were mostly found in Region I of case #7 and Region II for all cases. This composition range sat between the IMCs ($\gamma + \zeta$) and FCC-Cu (solid solution) regions (red arrow in Fig. 9a), with its typical microstructure being primary FCC-Cu mixed with IMCs, as shown in Fig. 9g. The IMCs phases present high hardness and brittleness, but their detrimental effects on the joints could be compromised by the soft and ductile FCC-Cu. The final microstructure was expected to present a good balance of strength and ductility.

3.3. Mechanical properties

Lap shear testing was performed for all nine cases, each with ten samples. The displacement control method at a loading rate of 1 mm/min guaranteed the same load on the two sides of the samples. The force versus displacement curves were recorded and plotted in Fig. 10.

Abrupt load drops were found in the load-displacement curves for cases #4, #5, and #7–#9, indicating brittle failure for these joints. Conversely, the curves in other cases were smooth, suggesting a gradual loss of joint strength. The failure modes of the joints were further investigated with the macro-scale images of the failed joints in Fig. 11. In cases #4, #5, #7–#9, cracks were found to initiate in and propagate through the fusion zone of the spiral welds. In the other cases, the cracks were found initiate in the base metal beyond the outside edge of the 3rd spiral arc and then propagate in the fusion zone. Detailed fractographic analyses were performed for cases #1, #3 and #7. The regions where the cracks were believed to initiate (marked out by boxes A, B, and C in Fig. 11) were selected for detailed characterization.

Fig. 12a showed the SEM image for the fracture surface of case #1. The field of view covered a small region on the top surface of the Cu plate (the yellow dashed boundary), a segment of fractured joint (the pink dashed boundary), and a small region of the top surface of the Al plate (blue dashed boundary). Based on the observation of Fig. 11, the joint failure started at the intersection between the outside edge of the fusion zone for the outmost spiral arc and the contact plane between Al and Cu in the lap joint, labeled by the red box (c). An enlarged SEM image of the area (Fig. 12c) exhibited parabolic dimples with their long axes parallel to the leading direction, which is consistent with the typical morphology for ductile fracture. The EDS results showed that the area was composed of pure Al, proving that the joint failure started in Al rather than the fusion zone. The crack then propagated across the fusion zone, for which the chemical composition ranged between 30–80 wt% Cu (as shown in the EDS mapping in Fig. 12b). Such composition corresponded to brittle IMC phases as shown in the Al-Cu phase diagram (Fig. 9a). Some cleavage morphology was observed in the area, indicating brittle fracture. The analysis suggested a mixed failure mode for the case #1 joint: the crack started in Al with a ductile mode, and

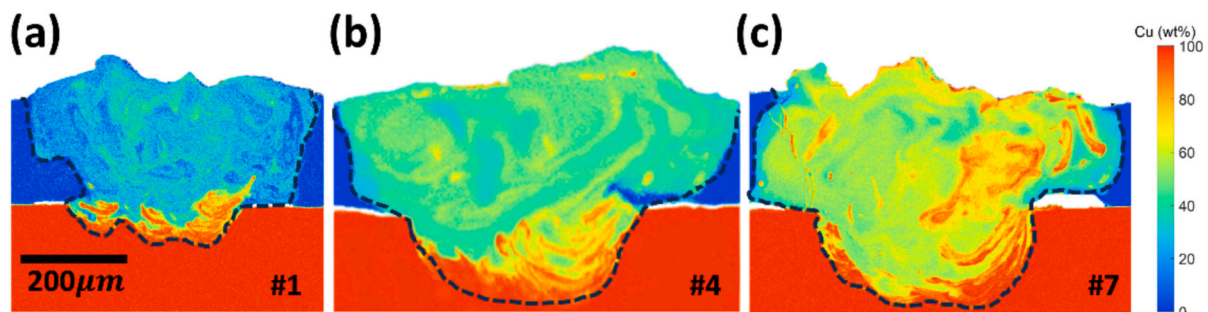


Fig. 8. EDS element mapping results of Cu concentration field for selected cases of different laser powers. (a) case #1 (b) case #4 (c) case #7.

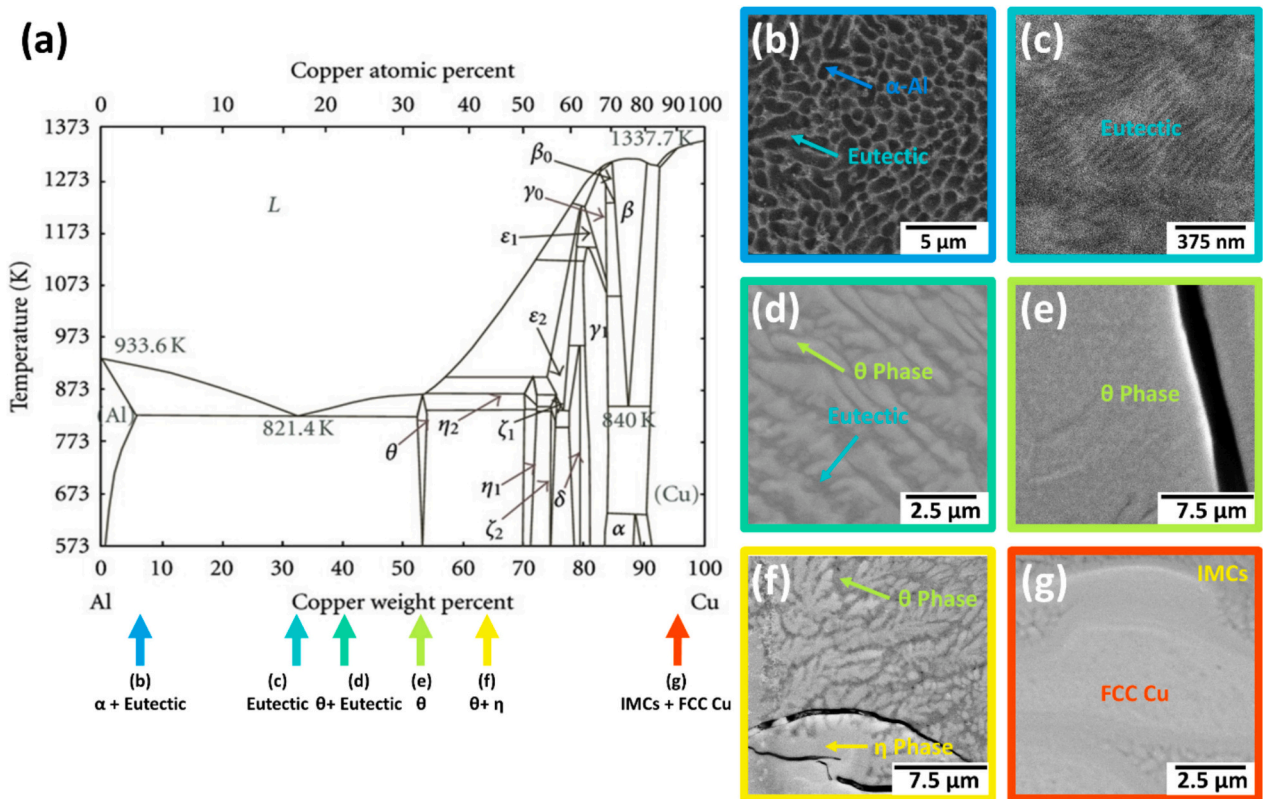


Fig. 9. Typical microstructures in the fusion zones. (a) Al-Cu phase diagram. (b) Hypo-eutectic phase. (c) Eutectic phase. (d) Hyper-eutectic phase. (e) θ phase. (f) $\theta + \eta$ mixture. (g) IMCs + FCC Cu.

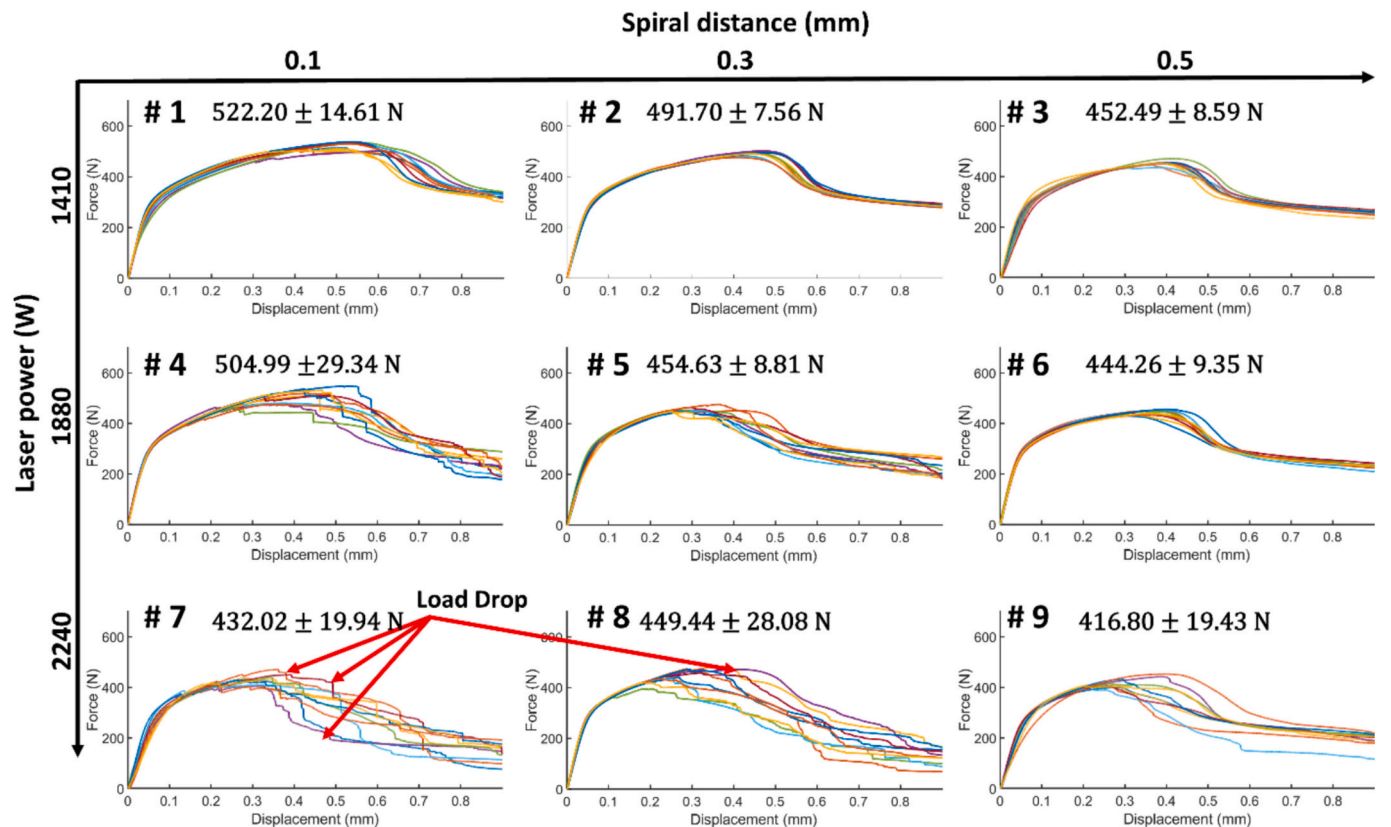


Fig. 10. Lap shear test: load-displacement curve and peak loads for the lap joints.

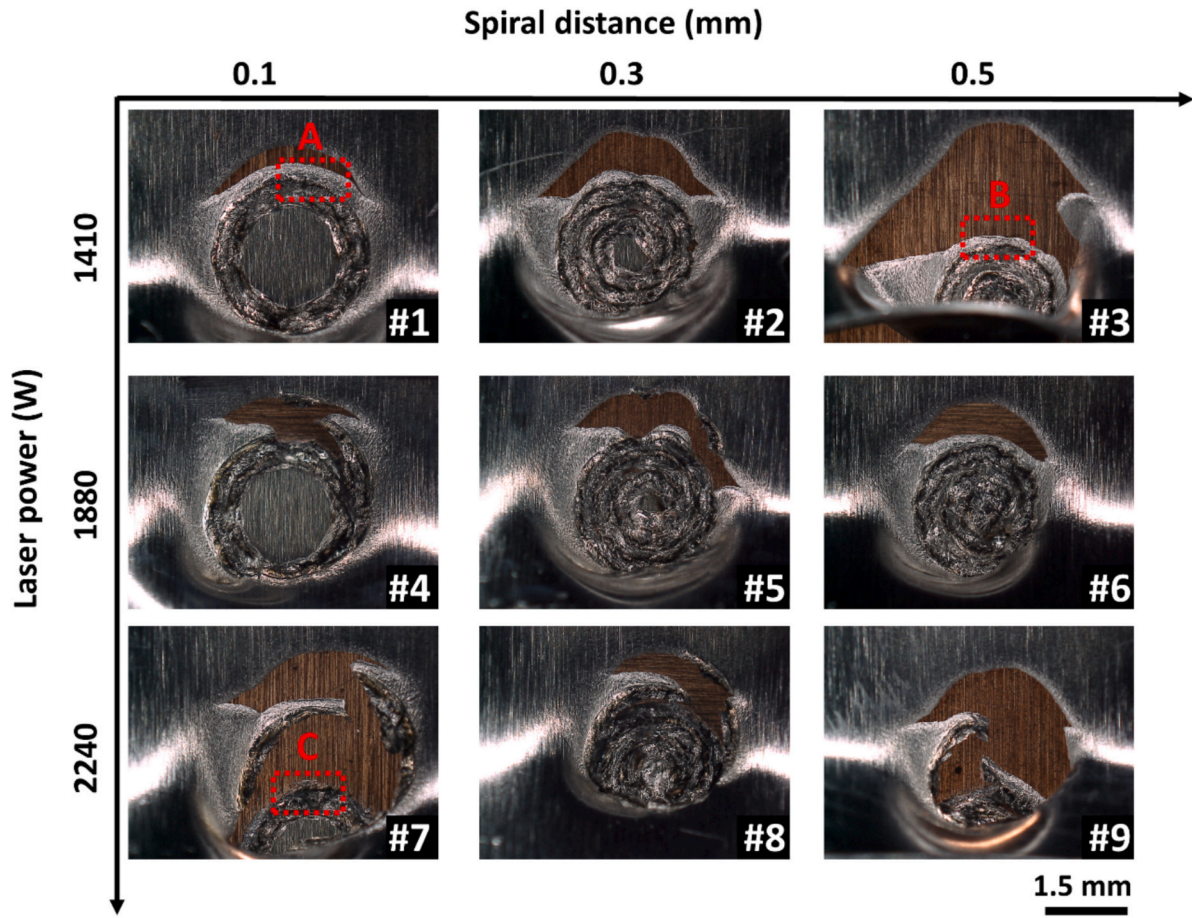


Fig. 11. The macroscale fractographic analysis of the Al-Cu spiral welds.

propagated in the fusion zone with a brittle mode. The detailed SEM and EDS images for the fracture surface of case #3 were shown in Fig. 12d–f, which presented similar information that led to similar conclusions.

Micro-scale fractographic analysis was also performed for case #7. In Fig. 12g, the joint failure was believed to start in the area marked out by the red box (i). The EDS mapping (Fig. 12h) showed a chemical composition range of 30–80 wt% Cu throughout the entire fracture surface, suggesting the existence of IMC phases. The zoom-in SEM image (Fig. 12i) presented visible cracks, notches, sharp edges/corners, all typical morphologies for brittle fracture. The case #7 joint failed within the fusion zone with an exclusively brittle mode. The joints for the other brittle cases (#4, #5, #8, and #9) were found to fail in very similar manners and presented similar fractographic morphologies.

As the joints for cases #1 and #3 started their crack in Al, a possibility was that the crack initiated in the heat affected zones (HAZ) but not the base metals. To check the existence of HAZ, microhardness tests were performed for the joints of cases #1, #3, and #7 (shown in Fig. 13). As the Cu concentration in the joints was raised from cases #3 to #1, and then to #7, the microstructures possessed growing percentages of IMC phases, causing the hardness in the joints to increase in these cases. On the other hand, the points outside the fusion zone were found to have low hardness values (HV32 ~ 36) similar to those for pure Al. The hardness drops from the fusion zone to the Al substrate were sharp, showing no apparent HAZ outside the fusion zones in all three cases. The cracks in cases #1 and #3 were believed to initiate in the base metal.

While the peak loads were measured and given in Fig. 9, it was not necessarily a fair metric to measure the joint strength as the total weld lengths were different for the cases of different spiral distances (see Fig. 2). The unit weld strength was calculated (i.e., F_u) by dividing the peak load of a curve, F_{peak} , by the total weld length, L_{weld} (values given in

Fig. 2), as given in Eq. (2).

$$F_u = \frac{F_{\text{peak}}}{L_{\text{weld}}} \quad (2)$$

For each case, the average unit weld strength load, $\bar{F}_u$, was calculated based on the F_u values of all nine curves of the case. The results are given in Table 2.

It showed that $\bar{F}_u$ became higher as the spiral distance was increased, which was attributed to two causes. The first was related to the micro-structure change. As discussed in Section 3.1, the average Cu concentration of the entire fusion zone, $\bar{C}_{\text{Cu}}$, became lower as the spiral distance was increased. The joint with 0.5 mm spiral distance was found to possess a lower percentage of IMC phases (of high Cu concentration) than the joint with 0.1 mm spiral distance, and therefore, presented to be less brittle. The second cause was related to the change of fusion zone geometry, or more specifically, the increase in the total width of fusion zone(s). When the spiral distance was 0.1 mm, the three consecutive fusion zones for the three spiral arcs were heavily overlapped to produce one integral fusion zone, the width of which was ~0.52 mm (Fig. 5a). When the spiral distance was 0.3 mm, the three consecutive fusion zones only had overlaps in their Region I, and the connected fusion zone had a total width of ~1.01 mm (Fig. 5b). When the spiral distance was 0.5 mm, the three consecutive fusion zones were completely separated. The width of each fusion zone was about 0.37 mm, and the total width of all fusion zones was about 1.40 mm. Gaps could be found in Fig. 5 between the Al and Cu plates in the fusion zones, indicating possible incomplete joining at certain locations, but the overall increasing trend for the total width of fusion zone(s) should be acknowledged, and this allowed the joints with larger spiral distances to distribute more mechanical loading over more space before the local stress level went beyond the material

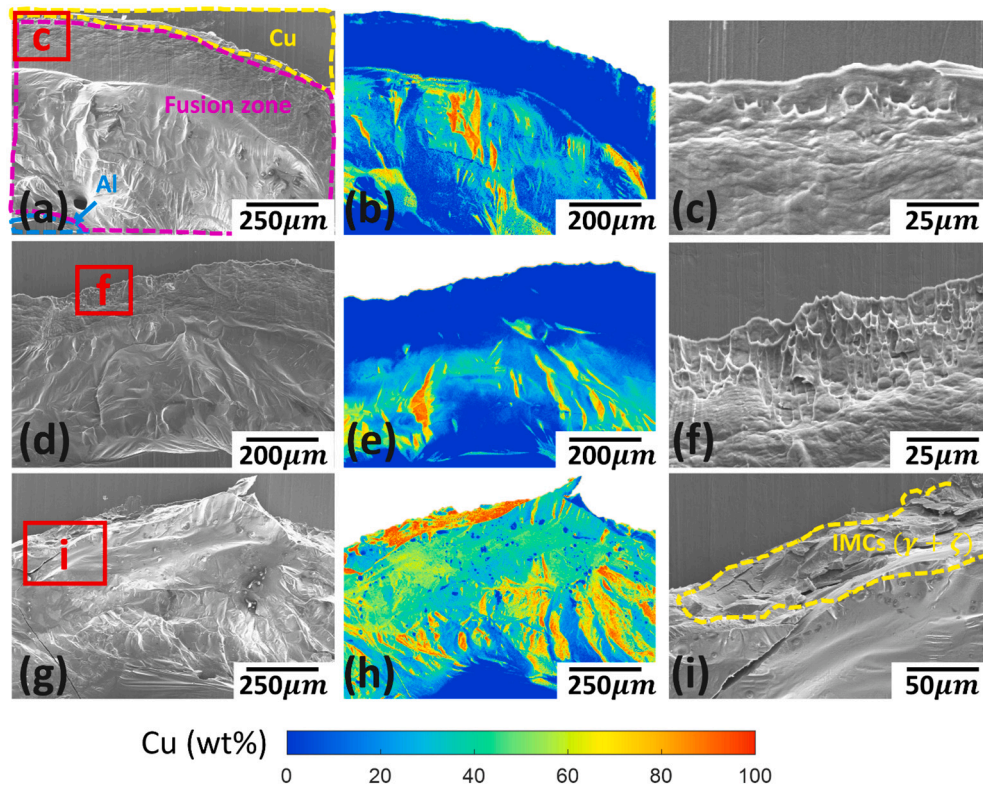


Fig. 12. SEM and EDS images for fractographic analyses of selected case. (a)-(c) are results for case #1, (d)-(f) are results for case #3, and (g)-(i) are results for case #7.

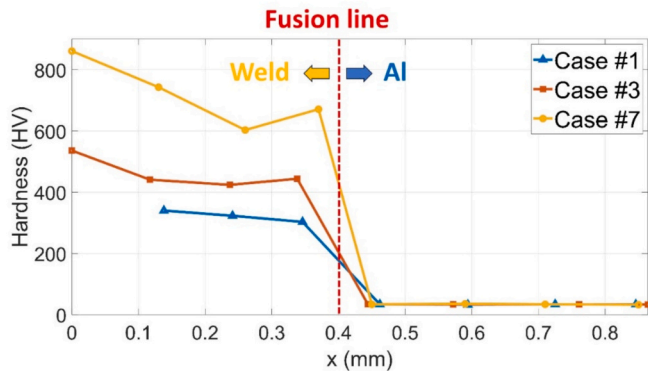


Fig. 13. Micro-hardness test results for selected cases.

Table 2

Normalized lap shear test result for all 9 cases.

Spiral distance	0.1 mm	0.3 mm	0.5 mm
Laser power			
1410 W	20.51 ± 0.57 N/mm	24.82 ± 0.38 N/mm	31.64 ± 0.60 N/mm
1880 W	19.84 ± 1.15 N/mm	22.94 ± 0.44 N/mm	31.06 ± 0.65 N/mm
2240 W	16.97 ± 0.78 N/mm	22.68 ± 1.41 N/mm	29.14 ± 1.35 N/mm

limit.

Table 2 also showed that $\bar{F}_u$ was reduced as the laser power was increased. This could be explained by comparing the results for cases #1, #4, and #7. As the laser power became higher, the Cu concentration and typical in Region I was increased changed from ~35 wt% (case #1), ~55 wt% (case #4), and then ~65 wt% (case #7), as shown in Fig. 7.

The dominating phases in Region I of these cases were also changed from hyper-eutectic (Fig. 9d) to θ (Fig. 9e) and then $\theta + \eta$ mixture (Fig. 9f). In the hyper-eutectic microstructure, a continuous eutectic network provided a strong and ductile backbone and the θ phase acted as an effective reinforcement, the whole microstructure was expected to behave as a strong yet ductile composite material, and thus the joint in case #1 presented the highest unit weld strength. In the other two cases, the joints were dominated by brittle IMC phases that easily cracked and thus failed at a lower mechanical load.

4. Conclusions

This work was focused on laser keyhole welding of dissimilar metals and aimed to understand the fundamental physics that governs the fluid flow, heat/solute transfer, microstructure evolution in the melt pool as well as the mechanical strength of the final joints. A combination of experiments and simulations was used in this work. Parametric experiments were conducted to identify the effects of processing parameters (i. e., spiral distance and laser power) on the chemical composition, microstructure, and mechanical strength of the joints. A computational fluid dynamics (CFD) model was also leveraged to reveal the fluid flow and metal mixing within the melt pool. The major findings of the work are summarized as follows.

Experimental results revealed that the processing parameters significantly affected the chemical compositions of the joints. A rise in the laser power led to an increasing amount of Cu to be melted in the melt pool and hence the higher Cu concentration in the final joints. A reduction of the spiral distance led to a reduction in the amount of Al to be melted in the melt pool and hence a higher Cu concentration in the final joints. The reduction of the spiral distance also helped to produce a more uniform Cu distribution in the joints.

The CFD simulations verified the two major physics that drove the fluid flow and metal mixing in the melt pool. The first was the recoil pressure at the keyhole bottom, which drove an upward flow to bring

the Cu from melt pool bottom to the top. The second was the Marangonian force on the melt pool surface, which drove flow vortices in the melt pool to stir and mix the Al-Cu liquid. In the cases of smaller spiral distance, a larger portion of the fusion zone for the previous spiral arc was re-melted and re-mixed by the melt pool of the next spiral arc, which effectively improved the homogeneity of Cu distribution.

The Cu distribution was not always homogeneous throughout the fusion zones. Instead, the fusion zones in most cases were shared by multiple regions, each with a distinct chemical composition range. Five different typical compositions were identified among all cases, and their corresponding microstructures were characterized as hypo-eutectic (0–10 wt% Cu), hyper-eutectic (30–40 wt% Cu), θ -Al₂Cu (50–60 wt %), θ + η -AlCu (60–70 wt% Cu), and FCC-Cu+ γ -Al₄Cu₉ (80–100 wt% Cu). As the spiral distance was decreased and/or the laser power was increased, the average Cu concentration of the entire fusion zone was increased, and an increasing percentage of the fusion zone area was found to be occupied by IMC phases of higher Cu concentrations.

Last but not least, the mechanical performance of the joints was evaluated with a lap shear test, and the unit weld strength was calculated to represent the average loading that could be sustained by a unit length of the weld. The unit weld strength was found to be higher in the cases with a large spiral distances and lower laser powers, which was attributed to (i) the reduction of IMC phases in the microstructure and (ii) the increase of total width in the fusion zones to distribute the loading.

The fractographic analysis revealed that the joints for these cases started the crack initiated in the Al base metal with a ductile mode and then propagated through the fusion zone with a brittle mode.

On the other hand, the unit weld strength was lower in the cases with smaller spiral distances and higher laser powers. The joints were found to crack exclusively with a brittle mode, with excessively IMC phases identified on the crack surface.

CRedit authorship contribution statement

Guanjin Yan: Writing – original draft, Validation, Software, Investigation, Formal analysis. **Masoud M. Pour:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Teresa J. Rinker:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Junjie Ma:** Writing – review & editing, Resources, Methodology, Investigation. **Blair E. Carlson:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Wenda Tan:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

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Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

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